

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1182

Walerga- Antelope 138 kV
May 03, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1182. The Interconnection Customer proposes a 111.9 MW wind facility interconnecting at the Walerga-Antelope 138 kV substation in the Virginia area. The request was received on April 06, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 452 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$206,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Lake Lure-Littlefork 138 kV	Fluvanna P.S-Louisa Generation Facility 345 kV	113.5	593	6.9
Lake Lure-Bluff City 138 kV	Fluvanna P.S-Louisa Generation Facility 345 kV	104.4	706	6.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lake Lure-Littlefork 138 kV	Rebuild line with higher-capacity conductor	\$6,471,000	2% (\$110,000)
Lake Lure-Bluff City 138 kV	Reconductor line	\$4,748,000	2% (\$96,000)
	Total allocated to project		\$206,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.